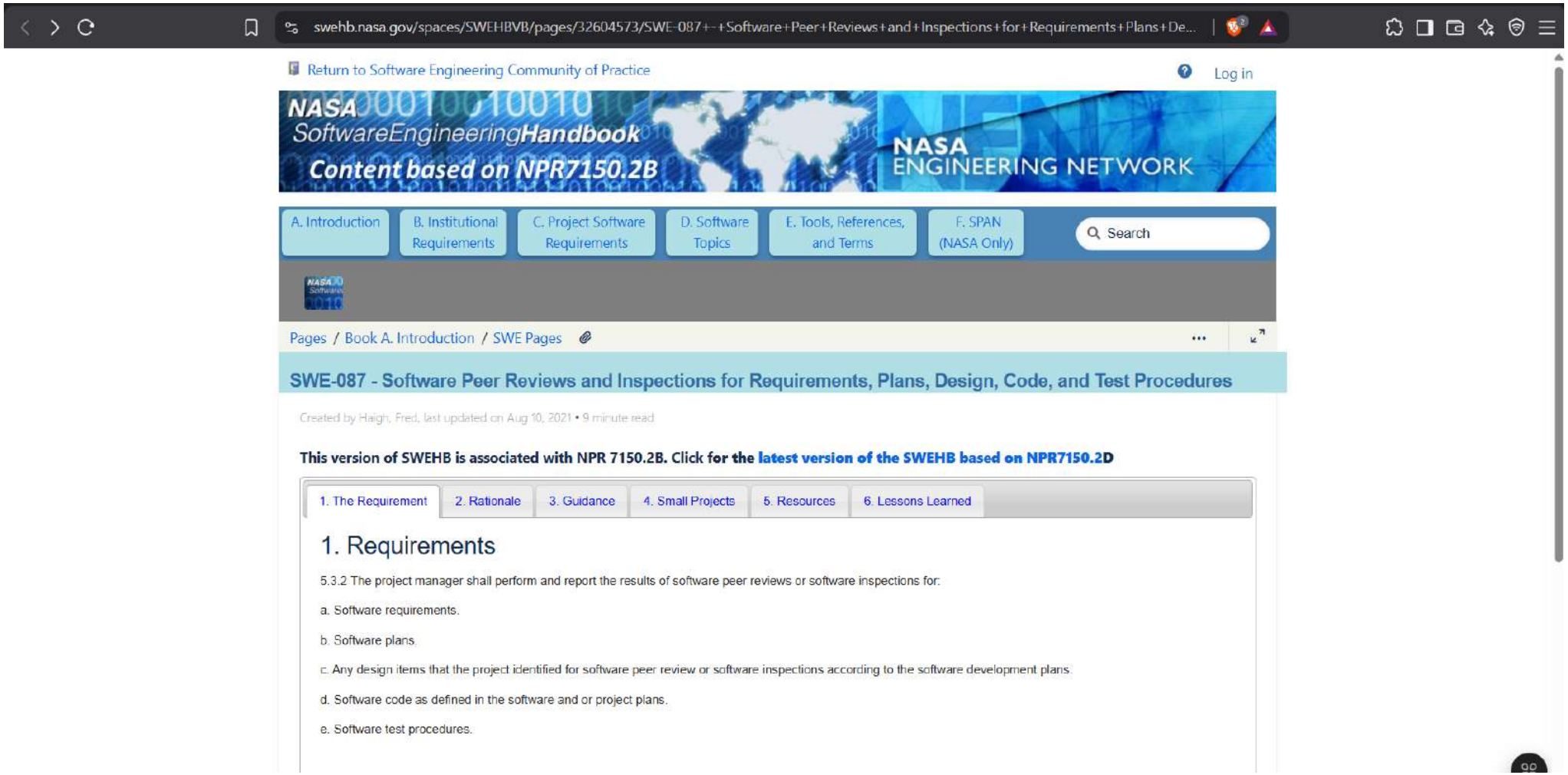

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SWE-087 - Software Peer Reviews and Inspections for Requirements, Plans, Design, Code, and Test Procedures
Created by Haigh, Fred, last updated on Aug 10, 2021 • 9 minute read
This version of SWEHB is associated with NPR 7150.2B. Click for the latest version of the SWEHB based on NPR7150.2D
1. The Requirement
2. Rationale
3. Guidance
4. Small Projects
5. Resources
6. Lessons Learned
1. Requirements
5.3.2 The project manager shall perform and report the results of software peer reviews or software inspections for:
a. Software requirements.
b. Software plans.
c. Any design items that the project identified for software peer review or software inspections according to the software development plans.
d. Software code as defined in the software and or project plans.
e. Software test procedures.



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1. Requirements

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1.1 Notes

Software peer reviews or software inspections are a recommended best practice for all safety and mission-success related software components. Recommended best practices and guidelines for software formal inspections are contained in NASA-STD-8739.9 [277].

Peer reviews/inspections provide the following additional benefits:

Useful for many types of products: documentation, requirements, designs, code	Simple to understand
Provide a way for sharing/learning of good product development techniques	Serve to bring together human judgment and analysis from diverse stakeholders in a constructive way
Can result in very efficient method of identifying defects early in the product's life cycle	Use a straight-forward, organized approach for evaluating a work product - To detect potential defects in a product To methodically evaluate each defect to identify solutions and track incorporation of these solutions into the product

Projects are required to conduct peer reviews/inspections for the following documentation based on software classification:

Software Documentation	A	B	C	D	E
Software Requirements	X	X	X	X (SC only)	
Software Plans	X	X	X	X (SC only)	
Software Design Identified in Plans	X	X	X	X (SC only)	
Software Code identified in Plans	X	X	X	X (SC only)	
Test Procedures	X	X	X	X (SC only)	

When conducting a peer review/inspection, be sure the following are part of the process in order to have an effective software peer review/inspection:

- Keep the review focused on the technical integrity and quality of the product.
- Keep the review simple and informal.
- Concentrate on review of the documentation and minimize presentations.
- Use a round-table format rather than a stand-up presentation.
- Give a full technical picture of items being reviewed.
- Plan the review/inspection, use checklists, and include readiness and completion criteria.
- Capture action items, monitor defects, results, and effort.

When putting a software peer review team together, use the following best practices:

- Team consists of minimum of three inspectors.
    - Diverse viewpoints and objectivity are required.
- Inspection team members are based on analysis of key stakeholders in the item under inspection.

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- Author should not be the reader, recorder, or moderator.
- Moderator should not be a manager.
- At a minimum, moderator should be formally trained in the process, but all participants may be trained.
- Management presence/participation discouraged.
- Each role has a specific responsibility as shown in the table below:

Role	Responsibility
Moderator	Conducts and controls the inspection
Author	Producer of product under inspection, answers technical questions
Reader	Presents (reads, paraphrases) the inspection product to the inspection team
Recorder	Documents defects identified during the inspection as well as open issues and action items

As documented in NASA-GB-8719.13, NASA Software Safety Guidebook [276], "Formal Inspections have the most impact when applied early in the life of a project, especially the requirements specification and definition stages of a project. Impact means that the defects are found earlier, when it's cheaper to fix them....Formal Inspection greatly improves the communication within a project and enhances understanding of the system while scrubbing out many of the major errors and defects."

Software peer reviews are conducted using the following steps:

Step	Description
Planning	Organize inspection, inspection package contents, required support, schedule
Overview	Educational briefing at time of package distribution to explain materials at high level
Preparation	Inspectors individually look for and document defects and develop questions
Inspection Meeting	Inspectors examine the product as a group to classify and record defects, capture open issues and action items
Third Hour	Optional informal meeting to resolve open issues and discuss solutions
Rework	Author corrects major defects (others when cost and schedule allow)
Follow-up	Moderator verifies all major and other dispositioned defects have been corrected; no new defects introduced; all action items/open issues are closed

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